

Samuel Xavier Borges

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science, Minor in Data Science, Minor in Math

Orlando, FL

Expected Grad: May, 2028

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

Unary Lab - University of Central Florida

Orlando, FL

- Characterized the **Llama-2-7B** inference workload for **Mugi**, an in-house **LLM accelerator**, sorting all **2,500+** per pass operations into **four** hardware execution buckets and emitting a validated **ONNX** graph for the cost/cycle simulator.
- Rearchitected the **Python/PyTorch** profiling pipeline around a **plugin adapter registry** that instruments any **Hugging Face** model without core changes, retiring per model if/else dispatch and fixing a latent binning format defect that had silently broken instrumentation for every model outside **one** hardcoded config.
- Consolidated per layer hand editing into a **single command** with a one shot **profiling API** that emits per layer lookup table configs and hardware workload specs from one config object through a pluggable **strategy/criterion sizer**, validated end to end against the simulator on the **NCSA Delta HPC** cluster.
- Cut a **32 round** manual search to **one** unattended job by building an in process **evaluation harness** that repatches lookup tables in place instead of reloading a **13GB** model per layer, sweeping a **16 window** grid at every layer seeded from measured exponent histograms.

Software Engineering Intern

Feb 2025 – Jan 2026

Knight Hacks

Orlando, FL

- Drove site **accessibility** to **WCAG 2.1 AA** keyboard navigation compliance with **15 reusable React components** replacing non semantic SVG elements with labeled, **ARIA** attributed alternatives.
- Delivered personalized acceptance, denial, and event detail email to **1,000+** applicants for one of **Florida's largest hackathons** via a **Google API** automation system built from scratch.
- Built a database backed **email queue service** and **admin console** with **four tier** priority scheduling, batch recipient validation, and date range blacklisting, retiring ad hoc one off send calls platform wide.

Software Engineering Intern

May 2025 – Aug 2025

Data-Enabled Photovoltaics

Orlando, FL

- Automated parsing and normalization of heterogeneous metadata across **five instrument sources** with an object oriented **Python** parser built on **Pandas** and **SQLite**, cutting manual preprocessing time by over **75%**.
- Defined and validated **FAIR** data model standards across **5** pipelines, integrating workflow outputs to preserve **metadata provenance** and make analyses reproducible from raw capture to database record.
- Unified **EXIF** and **TIFF** metadata from electroluminescence captures with binary **Sinton flash test** records into one normalized **SQLite** schema, adding per file **failure logging** that surfaced malformed data instead of dropping it.
- Documented the parser/outputter/database split with an extension guide and **D2** architecture diagrams, giving non programmer researchers a path to onboard new instrument types without touching pipeline code.

PROJECTS

Ctrl-ARM | *Python, Pandas, scikit-learn, Electron, React, Google ADK, Whisper*

- Won **1st at ShellHacks 2025's Microsoft AI Challenge** with a hands free computer control system fusing **EMG** signals and **IMU motion data** behind a multimodal **Electron + React** overlay.
- Engineered **Pandas ETL** pipelines over **200Hz** EMG/IMU streams to drive per user calibration and feature extraction, holding **~90%** classification accuracy across **12+** gestures through a **scikit-learn** decision tree.

Tariffix | *TypeScript, React, Vite, Tailwind CSS, MongoDB, Chrome Extension API, Gemini API, Cheerio*

- Enabled real time tariff insights on **100K+** product listings through a **Vite** powered **Chrome extension** backed by an AI scraping pipeline for metadata and **HTS** codes, winning **Bitcamp 2025's "Best use of MongoDB."**
- Achieved under **5% average error** against official U.S. duty rates by validating scraped listings against a custom **MongoDB** dataset and resolving **HTS** codes through a **Gemini** agent network.

ThisFlyPlaysDrums | *Python, PyTorch, MuJoCo, NumPy, Pandas, CUDA, Docker, Three.js*

- Taught a simulated fruit fly to play drums with its full **MaleCNS** connectome with **166,700** neurons & **24.7M** connections by running it as a spiking network in **PyTorch** that drives a **MuJoCo** fly body holding drumsticks.
- Achieved **0.29 F1** on held out grooves, **3x** a shuffled connectome control (**0.10**) and up from **0.004** untrained, by refitting **41K** connections onto **203** leg motor neurons with sign constrained **least squares** from **21** human takes.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, Java, TypeScript, JavaScript, SQL, HTML, CSS

Libraries & Frameworks: PyTorch, CUDA, MuJoCo, ONNX, Pandas, NumPy, scikit-learn, React, Next.js, Node.js, Three.js, tRPC, Drizzle, Tailwind CSS, Vite, Electron, Google ADK, Gemini API

Developer Tools: Git, Linux, Docker, PostgreSQL, SQLite, MongoDB, Turborepo, pnpm, Vercel, Unity, Figma, D2